

KILLAM AGDM, AB, Canada

WMO: 712280

Lat: 52.8475N

Lon: 111.8719W

Elev: 2213

StdP: 13.56

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-19.5	-32.2	1.0	-25.2	-25.9	1.4	-19.3	29.7	20.1	25.9	16.4	6.4	130	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	84.4	62.7	81.0	61.6	77.8	60.3	66.7	78.9	64.5	76.3	62.5	73.6	9.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	90.9	72.8	60.0	83.9	69.9	58.0	78.0	67.5	32.6	78.8	30.7	76.6	29.2	73.6	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.6	19.0	-33.3	90.3	6.2	4.1	-37.7	93.3	-41.4	95.7	-44.8	98.0	-49.3	101.1
			-33.7	71.5	6.0	2.3	-38.0	73.2	-41.6	74.5	-44.9	75.8	-49.3	77.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	11.0	11.0	23.0	38.5	51.0	58.2	62.3	59.9	51.3	38.8	22.5	11.1	
	DBStd	21.94	15.12	13.86	13.78	9.30	7.77	5.54	5.01	5.71	7.82	8.97	12.75	13.91	
	HDD50	6049	1209	1093	837	354	82	6	0	3	76	359	824	1206	
	HDD65	10390	1674	1513	1301	794	436	213	113	174	414	813	1274	1671	
	CDD50	1193	0	0	0	10	113	253	383	309	114	11	0	0	
	CDD65	59	0	0	0	0	2	10	30	15	2	0	0	0	
	CDH74	1641	0	0	0	5	133	219	591	479	208	6	0	0	
	CDH80	394	0	0	0	0	20	34	159	121	60	0	0	0	
Wind	WSAvg	8.9	8.9	8.5	9.2	10.3	10.3	9.4	7.9	7.6	8.4	9.2	8.7	8.4	
Precipitation	PrecAvg	16.1	0.8	0.5	0.7	1.2	1.7	2.9	2.8	2.2	1.3	0.7	0.7	0.7	
	PrecMax	21.5	2.4	0.9	1.7	2.8	4.1	4.3	5.9	4.9	3.3	1.9	2.4	1.2	
	PrecMin	10.3	0.2	0.1	0.0	0.2	0.4	0.7	0.7	0.2	0.0	0.2	0.1	0.1	
	PrecStd	3.3	0.5	0.3	0.4	0.6	0.9	1.0	1.2	0.9	0.9	0.4	0.5	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	41.9	56.0	72.6	82.7	83.9	89.0	88.2	86.9	73.6	55.5	42.9	
		MCWB	35.7	36.2	44.3	50.5	56.3	61.9	66.2	62.4	59.3	53.5	43.7	36.8	
	2%	DB	38.4	37.9	47.7	67.0	77.8	79.9	84.4	83.2	79.9	64.7	47.1	38.3	
		MCWB	34.2	33.4	39.8	48.3	54.4	60.1	65.7	62.3	57.5	49.6	39.1	34.0	
	5%	DB	35.2	34.7	42.8	60.5	73.2	76.1	80.5	79.4	74.3	59.0	42.1	34.1	
		MCWB	31.9	31.3	36.7	44.6	52.7	58.6	64.5	61.9	56.0	46.5	36.1	30.6	
	10%	DB	31.5	31.3	39.0	55.3	68.5	72.1	76.6	75.5	68.1	53.8	38.2	30.4	
		MCWB	28.6	28.6	34.5	42.0	50.3	56.9	62.9	61.0	53.2	43.9	33.9	27.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.4	36.7	45.7	51.6	58.6	65.8	71.0	68.7	61.6	54.5	44.5	37.2
			MCDB	40.3	41.5	54.5	71.2	77.4	78.8	82.1	78.0	80.4	70.8	52.9	41.8
2%		WB	34.4	33.7	40.3	48.4	55.8	62.8	67.9	65.6	58.6	50.7	40.1	34.1	
		MCDB	38.0	36.9	47.5	65.3	73.2	75.1	80.3	78.2	76.3	62.3	46.4	38.1	
5%		WB	32.1	31.5	36.9	45.6	53.9	60.3	65.7	63.4	56.5	47.6	36.6	30.8	
		MCDB	35.2	34.6	42.1	58.5	70.1	71.6	77.6	75.6	71.9	57.5	41.3	34.0	
10%		WB	28.8	28.5	34.7	43.1	51.8	58.5	63.7	61.4	54.3	44.6	34.2	27.7	
		MCDB	31.4	31.3	38.7	54.1	66.1	69.4	74.4	73.3	66.7	52.9	38.0	30.4	
Mean Daily Temperature Range	5% DB	MDBR	18.2	19.6	18.6	22.1	25.2	22.1	24.2	25.7	26.1	22.6	17.9	17.2	
		MCDBR	19.6	21.6	22.0	33.3	33.3	29.6	32.3	34.5	38.4	32.1	22.2	19.9	
		MCWBR	16.5	17.8	15.2	17.0	14.4	13.2	16.3	16.5	18.1	18.4	15.3	16.6	
	5% WB	MCDBR	20.1	20.9	21.1	29.6	30.0	25.1	28.9	29.6	34.0	30.5	20.8	19.7	
		MCWBR	17.3	17.4	15.0	15.8	13.4	12.6	15.8	15.9	16.7	17.8	14.9	16.7	
Clear-Sky Solar Irradiance	taub	0.264	0.285	0.298	0.332	0.352	0.367	0.362	0.375	0.317	0.282	0.256	0.242		
	taud	2.284	2.285	2.382	2.358	2.352	2.344	2.387	2.328	2.456	2.533	2.449	2.329		
	Ebn at Noon	232	262	285	286	285	280	280	268	275	266	240	223		
	Edh at Noon	19	26	30	35	36	37	35	35	27	20	16	15		
All-Sky Solar Radiation	RadAvg	335	668	1134	1497	1833	1903	1990	1623	1137	671	335	248		
	RadStd	32	52	86	140	124	130	102	122	121	87	35	23		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page